

Use-Case Flow Specification

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Use Case: Reserve Charging Slot

Primary Actor: EV Driver

Supporting Actors: Payment Gateway (external system)

Related Use Cases: «include» Check Connector Compatibility; hands off to Make Payment

Preconditions

1. The driver is logged in and has a registered vehicle profile (connector type).
2. The driver has searched for stations and selected a station with at least one socket.

Postconditions

Success: the selected socket is locked for the driver for the reserved window, a reservation record exists, and the driver has been handed off to Make Payment.

Failure: no reservation record is created and the socket remains available to other drivers.

Main Success Scenario

1. Driver selects a charging station and a specific socket from the search results.
2. System performs Check Connector Compatibility («include») to verify the vehicle's connector type matches the selected socket.
3. System displays the available 30-minute time slots for the selected socket.
4. Driver selects a reservation window.
5. System locks the socket for the driver and creates a reservation record for that window.
6. System calculates the applicable power rate for the reserved window and displays it to the driver.
7. System hands the driver off to Make Payment to confirm the reservation.
8. On successful payment, the system confirms the reservation and sends the driver a confirmation notification with a reservation ID.
9. Use case ends successfully.

Alternate Flows

2a. Connector Incompatible

1. System flags the selected socket as incompatible and displays an error message to the driver.
2. System returns the driver to the station search results to choose a compatible socket.
3. Use case resumes at step 1, or ends if the driver cancels the search.

5a. Slot No Longer Available (concurrent booking)

1. System detects that the selected slot was locked by another driver at the same instant.
2. System displays a "slot unavailable" message and refreshes the list of available slots.
3. Driver selects an alternate slot or station; use case resumes at step 3, or ends if the driver cancels.